STUDY OVERVIEW

CacheWise replication and KV cache recovery

Serving coding agents under GPU memory pressure

01

Replicate

Separate scheduling from eviction effects.

02

Recover

Compare recomputation with host restoration.

03

Schedule

Make request priority aware of host hits.

The same prefix is useful again

Model request
Prefill + decode

→

Tool runs
GPU state competes for space

→

Next request
Old prefix + new context

When the agent returns…

Reusable prefix | **+ new**

GPU eviction creates a recovery decision.

Recompute

Run prefill again.

Restore

Copy saved KV state back.

Replication separates two decisions

Scheduler × eviction policy

	LRU	Predictive
FCFS	**FCFS / LRU**	**FCFS / predictive**
Prefix-aware	**Prefix / LRU**	**Prefix / predictive**

Coverage study

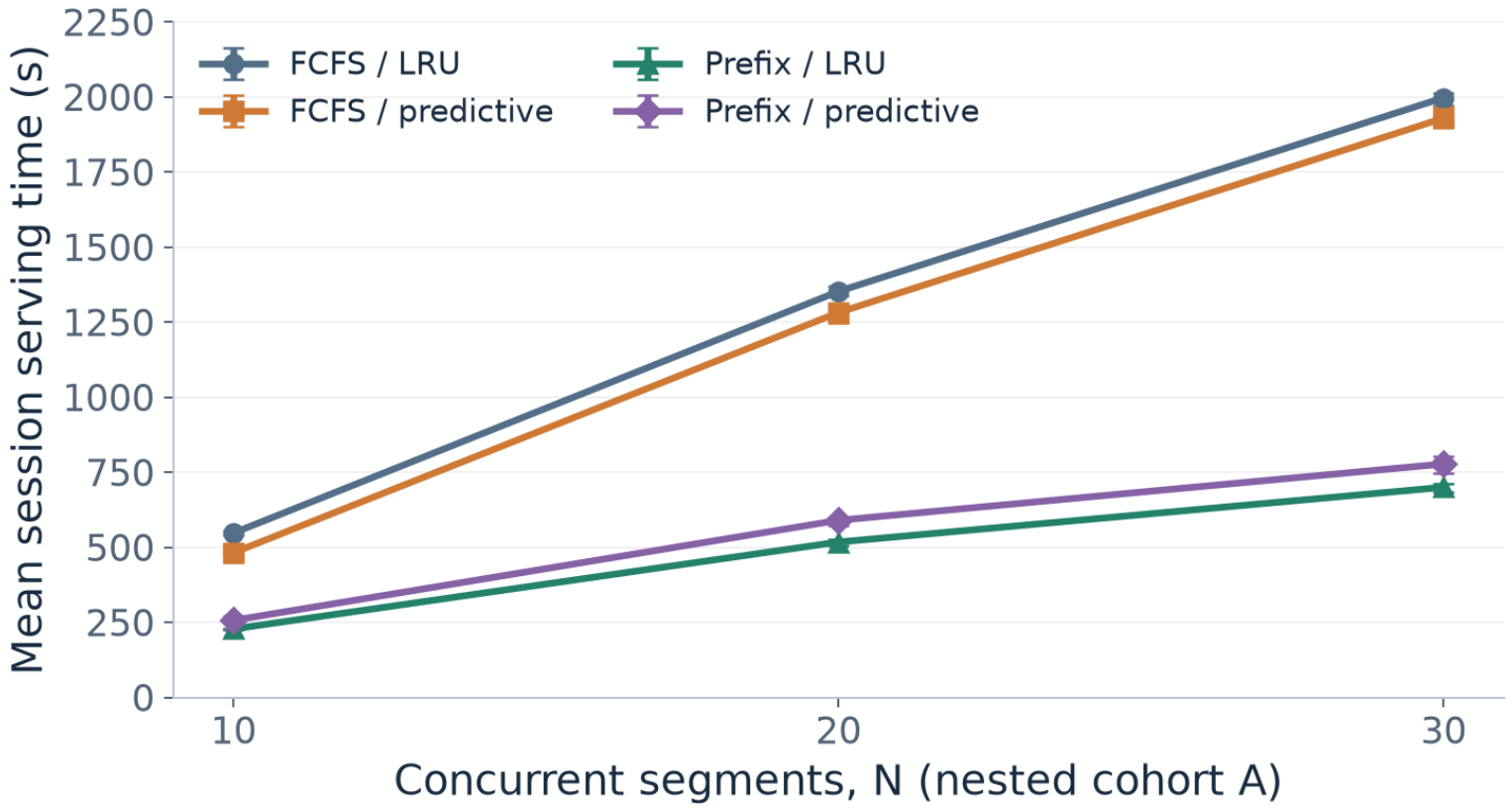
5 settings × 3 repeats
60 runs · 6,384 requests

14B model · BF16
287,696 KV token slots
512-token prefill chunks

Primary metric: mean session serving time, excluding tool waits.

Evidence: part1-coverage-comparison.json; report §3.1–3.2

Prefix / LRU has the lowest mean



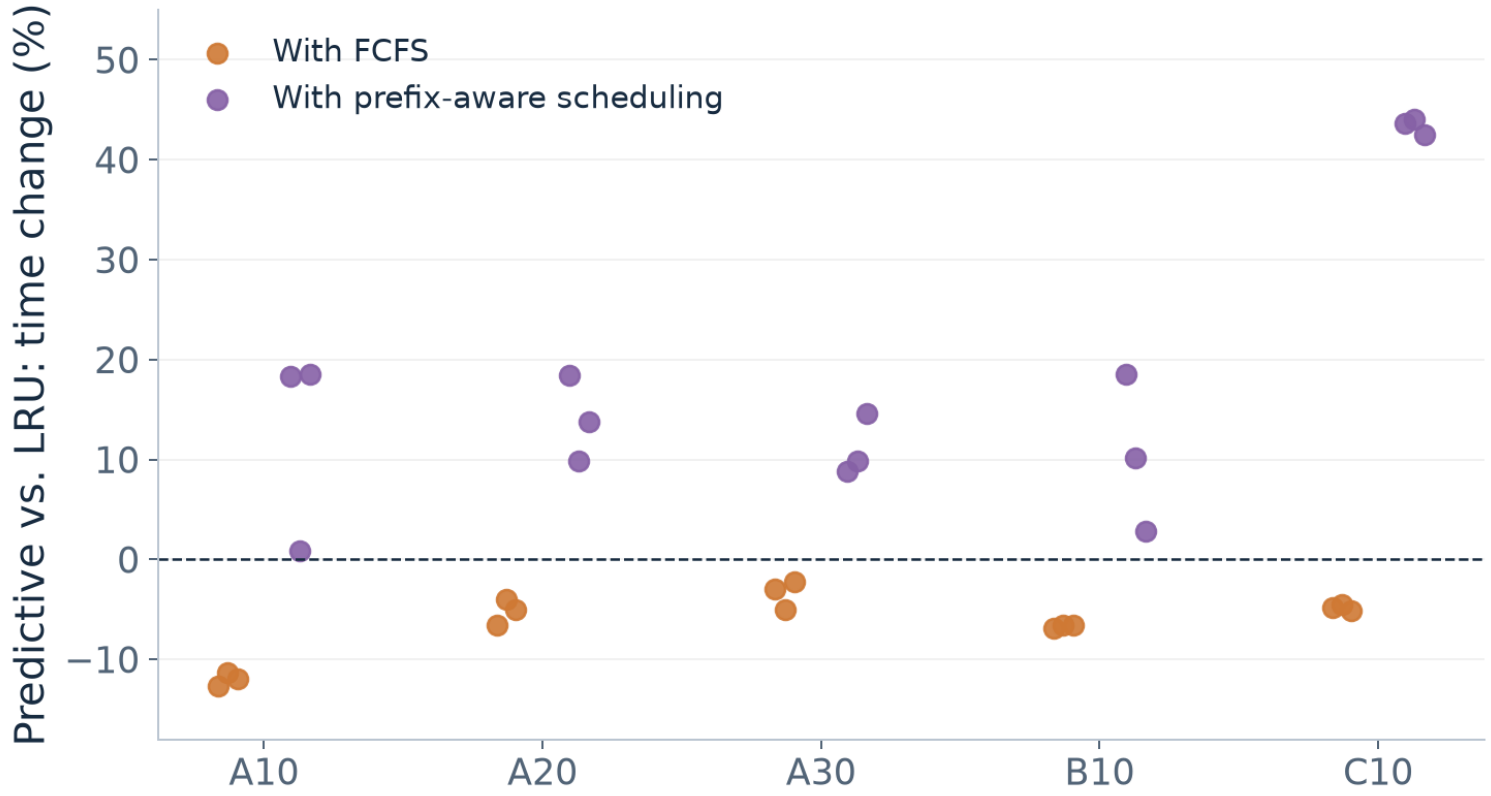
5 / 5

settings favor prefix / LRU

58–65% lower mean vs. FCFS / LRU in cohort A

Three repeats per cell. Ranges are observed minima/maxima; lower is better.

Prediction depends on the scheduler



Mean ≠ tail

A30 request P99
(average of 3 runs)

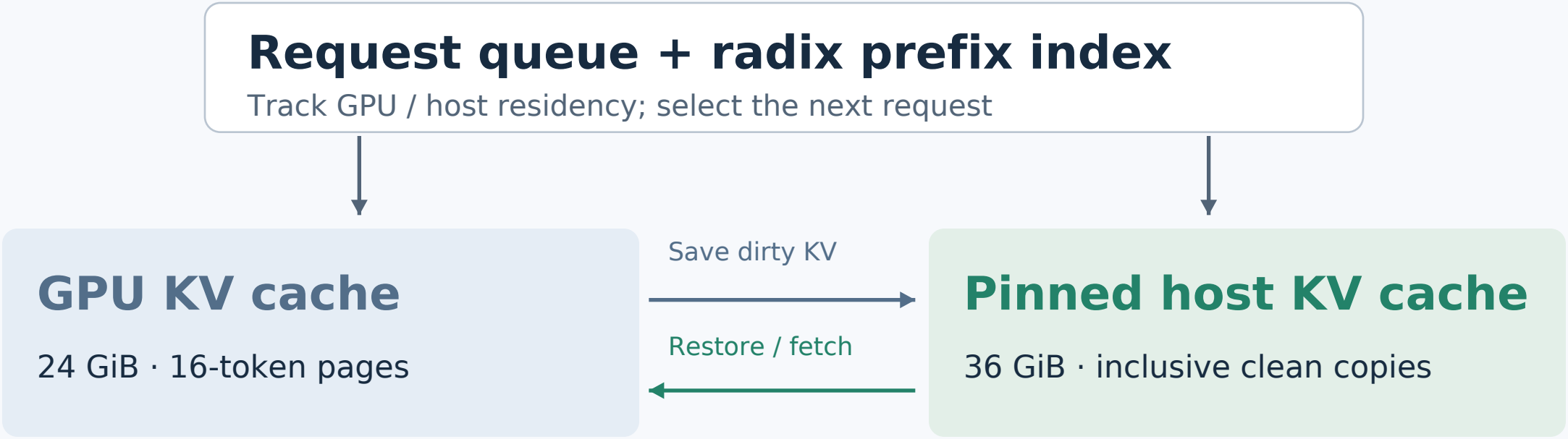
713 s → 1,292 s

FCFS / LRU →
prefix / LRU

Dots are matched-repeat effects. Positive values mean slower serving.

Evidence: matched coverage repeats and request tails; report Figure 1b and §3.3

Preserve evicted KV in host DRAM



On demand

Restore before prefill.

Prefetch

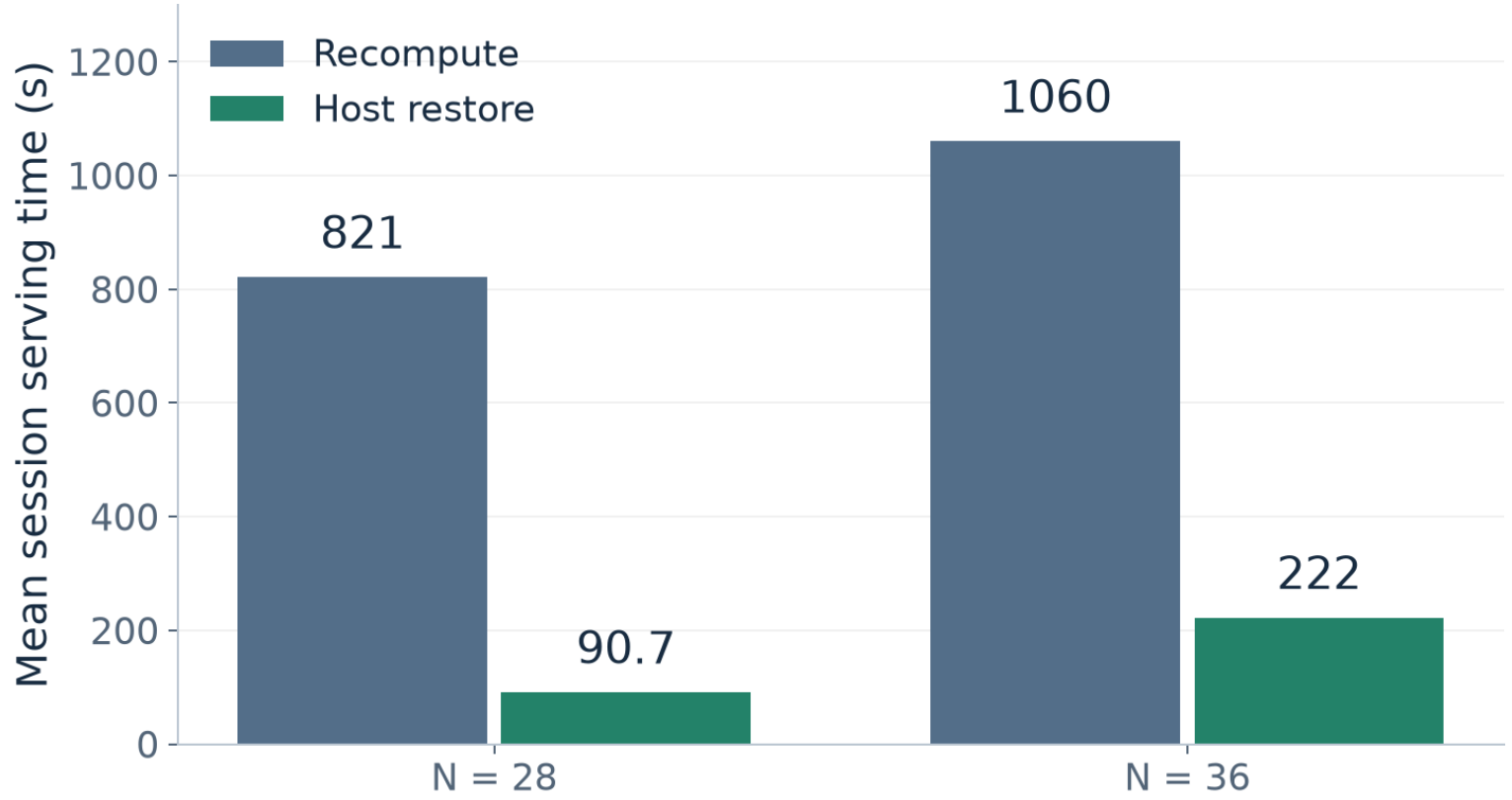
Overlap predicted reuse.

Fallback

Recompute on a miss.

Host restoration reduces session time

Recovery tradeoff: save + restore + management < recomputation



9.1× / 4.8×

reported time ratios
at N = 28 / 36

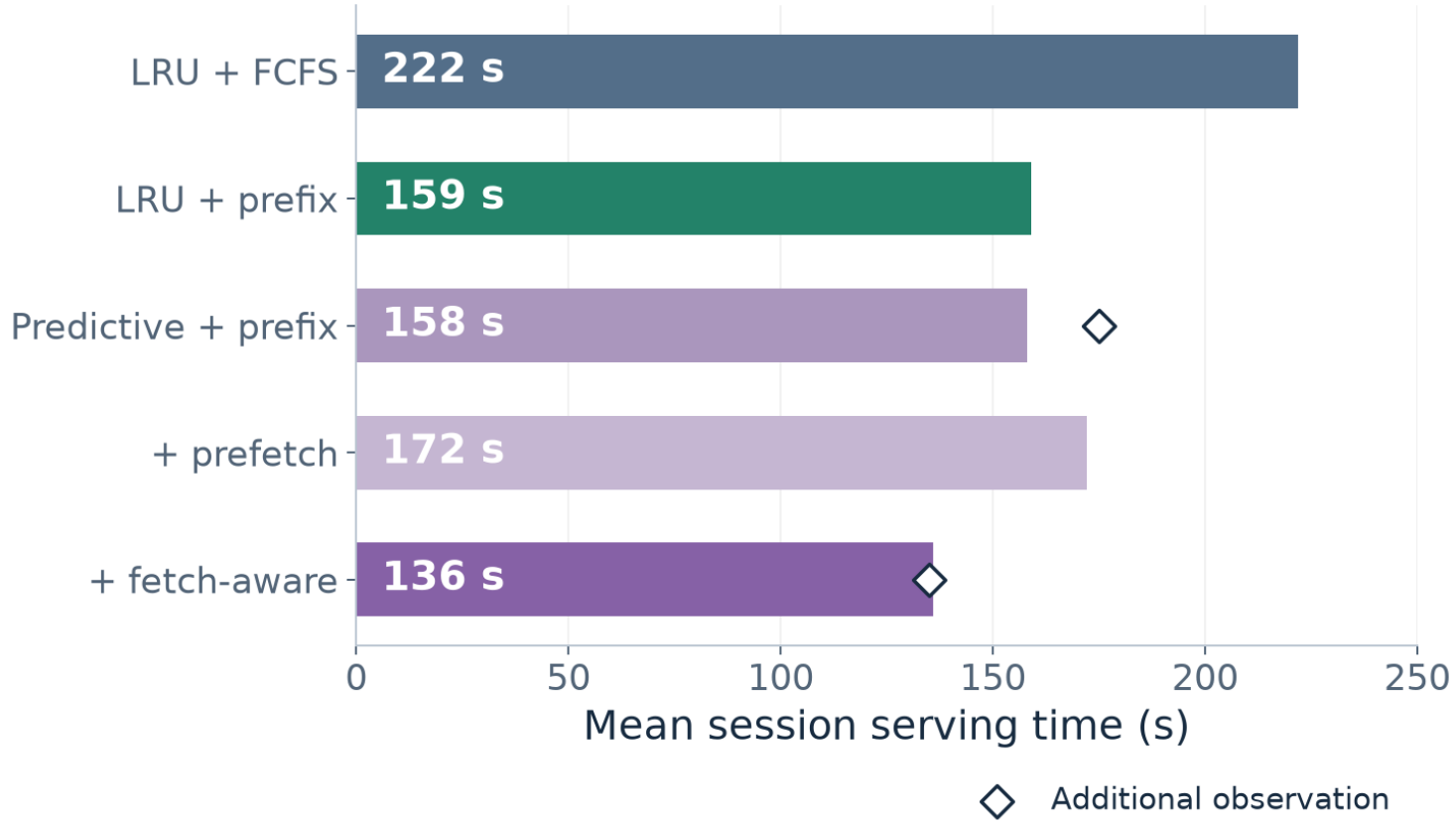
+36 GiB host KV
24 GiB GPU KV
fixed

One observation
per bar

Separate Mini-SGLang study. Added-capacity comparison; lower is better.

Evidence: host-dram-reported-results.json; report Figure 3a and §6

Fetch-aware ordering discounts host hits



Priority cost

new prefill tokens + 0.05 × host tokens

Mean of 2 observations
166.5 s → 135.5 s

Sparse repetitions; no confidence interval.

Bars: first observations. Diamonds: additional observations, not matched blocks.

Evidence: Mini-SGLangs 3f9aee7; report Figure 3b and Eq. (3)

Keep the conclusions within the evidence

Replication

Synthetic replay and selected segments.

One retained preemption; extra prefill work unmeasured.

Host results

Separate engine and replay.

Added host capacity; most configurations run once.

SSD extension

Not implemented or measured.

Needs full I/O, staging, metadata and recovery.

Interpret both studies separately; retain the declared protocol deviation.

Three takeaways

01 Scheduling and eviction interact.

Prefix / LRU has the lowest mean in all five replication settings.

02 Host copies can avoid costly recovery.

Large reported gains include the added 36 GiB backing tier.

03 Order requests by recovery cost.

Fetch-aware results are promising; SSD evaluation remains future work.